



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-294
JETERSVILLE-PONTON 115 KV
24.6 MW Capacity / 41 MW Energy**

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Dominion.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Amelia County, Virginia. The installed facilities will have a total capability of 41 MW with 24.6 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 01, 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-294
Project Name	JETERSVILLE-PONTON 115 KV
State	Virginia
County	Amelia
Transmission Owner	Dominion
MFO	41
MWE	41
MWC	24.6
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF1-294 will interconnect with the Dominion transmission system at a new 115 kV three breaker ring bus substation tapping the Jetersville to Ponton 115 kV line. Attachment 1 shows a one-line diagram of the proposed interconnection facilities. The IC may not install any facilities on Dominion's right-of-way without first obtaining the necessary approval from Dominion Energy.

5 Cost Summary

The AF1-294 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 8,800,000
Allocation towards System Network Upgrade Costs*	\$ 10,571,527
Total Costs	\$ 19,371,527

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start

with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Project AF1-294. The project was evaluated as a 24.6 MW Capacity (41 MW Energy) injection at a single line tap between Jetersville 115 kV substation and Ponton 115 kV substation in the Dominion Transmission System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2023 AF1 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically, in Planning Studies, NERC Planning Event 3 and 6 Contingency Conditions (Loss of generator, transmission circuit, transformer, shunt device, or Single Pole of a DC line followed by the loss of a generator, transmission circuit, transformer, shunt device or single pole of a DC line) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

The required Attachment Facilities, Direct Connection and Non-Direct Connection work for the interconnection of the AF1-294 generation project to the Dominion Transmission System is detailed in the

following sections. The associated one-line with the generation project attachment facilities and primary direct and non-direct connection are shown in Attachment 1.

Note that the ITO findings were made from a conceptual review of this project. A more detailed review of the connection facilities and their cost will be identified in a future study phases. Further note that the cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. ITO herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Attachment Facilities	\$ 1,700,000
Three breaker ring bus substation	\$ 5,500,000
Re-arrange transmission line to cut in new substation	\$ 1,600,000
Total Physical Interconnection Costs	\$ 8,800,000

It is estimated to take 24-30 months to complete this work. These preliminary cost estimates are based on typical engineering costs. A more detailed engineering cost estimates are normally done when the IC provides an exact site plan location for the generation substation during the Facility Study phase. These costs do not include CIAC Tax Gross-up.

Remote Terminal Work: During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

7 Schedule

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimate elapsed time to complete each of the required reinforcements is identified in the "System Reinforcements" section of the report.

8 Transmission Owner Analysis

8.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2023 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system and no further deficiencies were identified.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion’s “Dominion Energy Electric Transmission Generator Interconnection Requirements” documented in Dominion’s Facility Interconnection Requirements “Exhibit C” located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC’s GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion’s “Dominion’s Facility Interconnection Requirements” document located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.1.1 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit)
- Irradiance (Watts/meter²)
- Ambient air temperature (Fahrenheit) – (Accepted, not required)
- Wind speed (meters/second) – (Accepted, not required)
- Wind direction (decimal degrees from true north) – (Accepted, not required)

10.2 Interconnected Transmission Owner Requirements

See Section 3.4.6 “Metering and telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

11 Summer Peak Analysis

The Queue Project AF1-294 was evaluated as a 41.0 MW (Capacity 24.6 MW) injection tapping the Jetersville to Ponton 115 kV line in the Dominion area. Project AF1-294 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-294 was studied with a commercial probability of 100%. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
142108428	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P1-2: LN 556-A	single	571.52	99.39	101.12	AC	9.91
142108450	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P1-2: LN 556-A	single	559.30	98.62	100.39	AC	9.91

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
90243947	313867	6BREMODIST	230.0	DVP	313707	6FORK UNION	230.0	DVP	1	DVP_P1-2: LN 2027-B	single	661.76	111.54	112.54	AC	6.64
90243948	313867	6BREMODIST	230.0	DVP	313707	6FORK UNION	230.0	DVP	1	314333 6POWHATN 230 933500 AC2-165 TAP 230 1	single	661.76	111.53	112.53	AC	6.64
90243411	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P4-2: 511T591	breaker	699.0	107.8	110.15	AC	16.53
90243413	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P4-2: 556T591	breaker	699.0	107.8	110.15	AC	16.53
142108013	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P4-2: 511T556-A	breaker	699.0	107.8	110.15	AC	16.53
142108427	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P1-2: LN 556-B	single	571.52	101.7	103.44	AC	9.91
90243426	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P4-2: 511T591	breaker	684.0	106.52	108.92	AC	16.53
90243428	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P4-2: 556T591	breaker	684.0	106.51	108.92	AC	16.53
142108023	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P4-2: 511T556-A	breaker	684.0	106.52	108.92	AC	16.53
142108449	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P1-2: LN 556-B	single	559.30	101.04	102.82	AC	9.91
42958949	314702	3KERR	115.0	DVP	304102	3GW KING TAP	115.0	CPL	1	DVP_P4-6: CAROLIN T122	breaker	199.0	118.63	120.73	AC	4.95
42958950	314702	3KERR	115.0	DVP	304102	3GW KING TAP	115.0	CPL	1	DVP_P4-2: 2202	breaker	199.0	110.47	112.35	AC	4.33
42958951	314702	3KERR	115.0	DVP	304102	3GW KING TAP	115.0	CPL	1	DVP_P4-2: 13002	breaker	199.0	107.8	109.69	AC	4.33
42958952	314702	3KERR	115.0	DVP	304102	3GW KING TAP	115.0	CPL	1	DVP_P4-2: 101002	breaker	199.0	107.8	109.68	AC	4.33

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE T LOADIN G %	POST PROJE T LOADIN G %	AC/D C	MW IMPAC T
42959331	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P7-1: LN 22-90-A	tower	199.0	123.95	126.02	AC	4.89
42959332	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P7-1: LN 22-90-B	tower	199.0	112.66	114.74	AC	4.89
43252150	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P4-2: 102802	breaker	199.0	107.8	109.68	AC	4.33
90243927	31474 7	6BREMO	230.0	DVP	31386 7	6BREMODIS T	230.0	DVP	1	DVP_P1-2: LN 2027-B	single	661.76	113.83	114.84	AC	6.64
90243928	31474 7	6BREMO	230.0	DVP	31386 7	6BREMODIS T	230.0	DVP	1	314333 6POWHAT N 230 933500 AC2-165 TAP 230 1	single	661.76	113.82	114.82	AC	6.64

11.4 Steady-State Voltage Requirements

To be determined

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE T LOADIN G %	POST PROJE T LOADIN G %	AC/D C	MW IMPAC T
90243879	31370 7	6FORK UNION	230.0	DVP	31476 5	6MTEAGLE	230.0	DVP	1	DVP_P1 -2: LN 2027-B	operation	661.76	125.4	126.94	AC	10.12
90243943	31386 7	6BREMODIS T	230.0	DVP	31370 7	6FORK UNION	230.0	DVP	1	DVP_P1 -2: LN 2027-B	operation	661.76	122.48	124.17	AC	11.07
14210842 5	31467 7	6BUCKING	230.0	DVP	31474 7	6BREMO	230.0	DVP	1	DVP_P1 -2: LN 556-B	operation	571.52	129.06	131.92	AC	16.52
14210844 7	31469 2	6FARMVIL	230.0	DVP	31467 7	6BUCKING	230.0	DVP	1	DVP_P1 -2: LN 556-B	operation	559.30	127.4	130.34	AC	16.52
14210849 3	31469 7	6SEGE HILL	230.0	DVP	92725 0	AC1-221 TAP	230.0	DVP	1	DVP_P1 -2: LN 556-B	operation	674.92	125.78	126.98	AC	9.61
43252473	31470 1	3LONEPN	115.0	DVP	31470 7	3MORAN	115.0	DVP	1	DVP_P1 -2: LN 98-C	operation	203.98	83.75	102.79	AC	40.98
42959209	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P1 -2: LN 296-B	operation	199.0	105.29	106.98	AC	4.01

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
42959210	314702	3KERR	115.0	DVP	304102	3GW KING TAP	115.0	CPL	1	DVP_P1-2: LN 296-A	operation	199.0	105.28	106.97	AC	4.01
43252479	314707	3MORAN	115.0	DVP	314691	3FARMVIL	115.0	DVP	1	DVP_P1-2: LN 98-C	operation	203.98	81.68	100.73	AC	40.98
90243923	314747	6BREMO	230.0	DVP	313867	6BREMODIS T	230.0	DVP	1	DVP_P1-2: LN 2027-B	operation	661.76	124.79	126.47	AC	11.07
90243909	314765	6MTEAGLE	230.0	DVP	314749	6CHARLV	230.0	DVP	1	DVP_P1-2: LN 2027-B	operation	661.76	122.15	123.69	AC	10.12
42959093	927250	AC1-221 TAP	230.0	DVP	304070	6PERSON230 T	230.0	CPL	1	DVP_P1-2: LN 556-B	operation	718.0	130.19	131.3	AC	9.61

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-294	Upgrade Number																								
90243948,90243947	3	6BREMODIST 230.0 kV - 6FORK UNION 230.0 kV Ckt 1	<u>DVP</u> ProjectId : b2989 Description : PJM Baseline Upgrade b2989: Install a second 230 -115 kV Transformer(224 MVA) approximately 1 mile north of Brema and tie 230 kV Line #2028(Brema Charlottesville) and 115 kV Line #91 (Brema-Sherwood) together. A three breaker 230 kV ring bus will split Line #2028 into two lines and Line #91 will also be split into two lines with a new three breaker 115 kV ring bus. Install a temporary 230-115 kV transformer at Brema substation for the interim until the new substation is complete. The baseline project had an actual in-service date of 04/15/2020. Type : CON Total Cost : \$0 ProjectId : n6185.1 Description : Rebuild 1.63 miles of 230 kV Line 2028 from Fork Union to Brema Dist with 2-636 ACSR. Type : FAC Total Cost : \$2,445,000 Time Estimate : 30-36 Months Ratings : 1047.0/1047.0/1204.0 Queue Project AF1-294 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-294 could receive cost allocation. Note 2: Although Queue Project AF1-294 may not have cost responsibility for this upgrade, Queue Project AF1-294 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-294 comes into service prior to completion of the upgrade, Queue Project AF1-294 will need an interim study	\$2,445,000	\$0	b2989, n6185.1																								
90243413,142108427,90243411,142108013,142108428	1	6BUCKING 230.0 kV - 6BREMO 230.0 kV Ckt 1	<u>DVP</u> ProjectId : n6219 (dom-034) Description : Rebuild 15.51 miles of 230 kV Line 298 from Buckingham to Brema with 2-636 ACSR. Type : FAC Total Cost : \$23,265,000 Time Estimate : 30-36 Months Ratings : 1047.0/1047.0/1204.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-259</td><td>5.2</td><td>5.05%</td><td>\$1,175,114</td></tr><tr><td>AE2-291</td><td>11.14</td><td>10.82%</td><td>\$2,517,456</td></tr><tr><td>AE2-292</td><td>13.87</td><td>13.47%</td><td>\$3,134,391</td></tr><tr><td>AF1-059</td><td>15.67</td><td>15.22%</td><td>\$3,541,161</td></tr><tr><td>AF1-246</td><td>29.5</td><td>28.65%</td><td>\$6,666,513</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-259	5.2	5.05%	\$1,175,114	AE2-291	11.14	10.82%	\$2,517,456	AE2-292	13.87	13.47%	\$3,134,391	AF1-059	15.67	15.22%	\$3,541,161	AF1-246	29.5	28.65%	\$6,666,513	\$23,265,000	\$6,666,513	n6219
Queue	MW	Cost %	Cost \$																											
AE2-259	5.2	5.05%	\$1,175,114																											
AE2-291	11.14	10.82%	\$2,517,456																											
AE2-292	13.87	13.47%	\$3,134,391																											
AF1-059	15.67	15.22%	\$3,541,161																											
AF1-246	29.5	28.65%	\$6,666,513																											

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-294	Upgrade Number																												
90243426,142108023,142108449,90243428,142108450	2	6FARMVIL 230.0 kV - 6BUCKING 230.0 kV Ckt 1	<u>DVP</u> ProjectId : n6245 (dom-035) Description : Rebuild 12.76 miles of 230 kV Line 298 from Buckingham to Farmville with 2-636 ACSR. Type : FAC Total Cost : \$19,140,000 Time Estimate : 30-36 Months Ratings : 1047.0/1047.0/1204.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-291</td><td>5.07</td><td>6.26%</td><td>\$1,197,726</td></tr><tr><td>AE2-292</td><td>13.87</td><td>17.12%</td><td>\$3,276,621</td></tr><tr><td>AF1-059</td><td>5.01</td><td>6.18%</td><td>\$1,183,552</td></tr><tr><td>AF1-246</td><td>29.51</td><td>36.42%</td><td>\$6,971,382</td></tr><tr><td>AF1-293</td><td>11.03</td><td>13.61%</td><td>\$2,605,705</td></tr><tr><td>AF1-294</td><td>16.53</td><td>20.40%</td><td>\$3,905,014</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-291	5.07	6.26%	\$1,197,726	AE2-292	13.87	17.12%	\$3,276,621	AF1-059	5.01	6.18%	\$1,183,552	AF1-246	29.51	36.42%	\$6,971,382	AF1-293	11.03	13.61%	\$2,605,705	AF1-294	16.53	20.40%	\$3,905,014	\$19,140,000	\$3,905,014	n6245
Queue	MW	Cost %	Cost \$																															
AE2-291	5.07	6.26%	\$1,197,726																															
AE2-292	13.87	17.12%	\$3,276,621																															
AF1-059	5.01	6.18%	\$1,183,552																															
AF1-246	29.51	36.42%	\$6,971,382																															
AF1-293	11.03	13.61%	\$2,605,705																															
AF1-294	16.53	20.40%	\$3,905,014																															
90243927,90243928	5	6BREMO 230.0 kV - 6BREMODIST 230.0 kV Ckt 1	<u>DVP</u> ProjectId : n5873 Description : Rebuild 0.16 miles of Line 2028 from Brema to Brema Dist with 2-636 ACSR. Type : FAC Total Cost : \$500,000 Time Estimate : 30-36 Months Ratings : 1047.0/1047.0/1204.0 Queue Project AF1-294 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-294 could receive cost allocation. Note 2: Although Queue Project AF1-294 may not have cost responsibility for this upgrade, Queue Project AF1-294 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-294 comes into service prior to completion of the upgrade, Queue Project AF1-294 will need an interim study	\$500,000	\$0	n5873																												

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-294	Upgrade Number
42959331,42959332,42958949,42958952,42958950,42958951,43252150	4	3KERR 115.0 kV - 3GW KING TAP 115.0 kV Ckt 1	DVP ProjectId : n6115 (dom-002) Description : For DEV portion, rebuild 4.7 miles of 115 kV Line 45 from Kerr Dam to GW King Tap with 768 ACSS. Type : FAC Total Cost : \$6,123,000 Time Estimate : 30-36 Months Ratings : 393.0/393.0/452.0 Queue Project AF1-294 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-294 could receive cost allocation. Note 2: Although Queue Project AF1-294 may not have cost responsibility for this upgrade, Queue Project AF1-294 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-294 comes into service prior to completion of the upgrade, Queue Project AF1-294 will need an interim study CPL&E Note: Duke Energy/Progress Portion: A potential constraint was identified by PJM on the Duke Energy/Progress (DEP) portion of the Kerr Dam – GW King Tap 115 kV line. There are no mitigations currently planned for the DEP portion of this overload. The Queue Project AF1-294 may be subject to operational restriction if real-time system reliability issues occur. Additionally, if a baseline Network Upgrade project is identified on the Kerr Dam – GW King Tap 115 kV line prior to the execution of the Queue Project's final agreements, the Queue Project may require this upgrade to be in-service to be deliverable to the PJM system. If Queue Project AF1-294 comes into service prior to completion of the baseline Network Upgrade, Queue Project AF1-294 will need an interim deliverability study. Queue Project AF1-294 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-294 could receive cost allocation. Note 2: Although Queue Project AF1-294 may not have cost responsibility for this upgrade, Queue Project AF1-294 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-294 comes into service prior to completion of the upgrade, Queue Project AF1-294 will need an interim study	\$6,123,000	\$0	n6115
TOTAL COST				\$51,473,000	\$10,571,527	

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such

as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
142108013	314677	6BUCKING	DVP	314747	6BREMO	DVP	1	DVP_P4-2: 511T556-A	breaker	699.0	107.8	110.15	AC	16.53

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.2285	50/50	0.2285
314429	3JTRSVLE	0.9139	50/50	0.9139
314677	6BUCKING	1.6690	50/50	1.6690
315153	1CLOVER1	23.0700	50/50	23.0700
315154	1CLOVER2	22.8400	50/50	22.8400
924021	AB2-043 C O1	0.7651	50/50	0.7651
924022	AB2-043 E O1	3.8627	50/50	3.8627
924032	AB2-045 E	4.3976	50/50	4.3976
924161	AB2-060 C OP	2.2566	50/50	2.2566
924162	AB2-060 E OP	5.9958	50/50	5.9958
924301	AB2-077 C O1 (Suspended)	2.6930	50/50	2.6930
924302	AB2-077 E O1 (Suspended)	1.7954	50/50	1.7954
924311	AB2-078 C O1 (Suspended)	2.6930	50/50	2.6930
924312	AB2-078 E O1 (Suspended)	1.7954	50/50	1.7954
924321	AB2-079 C O1 (Suspended)	2.6930	50/50	2.6930
924322	AB2-079 E O1 (Suspended)	1.7954	50/50	1.7954
925611	AC1-036 C	1.8364	50/50	1.8364
925612	AC1-036 E	2.9963	50/50	2.9963
925781	AC1-054 C O1	5.1726	Adder	6.09
925782	AC1-054 E O1	2.3829	Adder	2.8
925831	AC1-062	0.1357	50/50	0.1357
925991	AC1-075 C	4.2230	Adder	4.97
925992	AC1-075 E	2.3927	Adder	2.81
926021	AC1-080 C	1.4114	Adder	1.66
926022	AC1-080 E	0.7939	Adder	0.93
926271	AC1-105 C O1 (Suspended)	6.7732	50/50	6.7732
926272	AC1-105 E O1 (Suspended)	3.3717	50/50	3.3717
926451	AC1-116 C	-0.3879	Adder	-0.46
927251	AC1-221 C	1.4232	Adder	1.67
927252	AC1-221 E	1.4232	Adder	1.67
927261	AC1-222 C	3.4595	Adder	4.07
927262	AC1-222 E	3.2933	Adder	3.87
932511	AC2-071 C	9.2838	50/50	9.2838

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
932512	AC2-071 E	4.6768	50/50	4.6768
932761	AC2-100 C	3.2753	Adder	3.85
932762	AC2-100 E	1.5986	Adder	1.88
934311	AD1-055 C	2.4020	Adder	2.83
934312	AD1-055 E	0.6194	Adder	0.73
934341	AD1-058 C	3.5677	Adder	4.2
934342	AD1-058 E	0.9065	Adder	1.07
934611	AD1-087 C O1	12.0586	50/50	12.0586
934612	AD1-087 E O1	5.6673	50/50	5.6673
934621	AD1-088 C	35.2780	50/50	35.2780
934622	AD1-088 E	16.6014	50/50	16.6014
934991	AD1-131 C	1.1697	Adder	1.38
934992	AD1-131 E	0.7798	Adder	0.92
935171	AD1-152 C O1	11.9837	50/50	11.9837
935172	AD1-152 E O1	7.9891	50/50	7.9891
935221	AD1-157 C	3.8762	50/50	3.8762
935222	AD1-157 E	2.5841	50/50	2.5841
936261	AD2-033 C	23.8095	50/50	23.8095
936262	AD2-033 E	15.8730	50/50	15.8730
936361	AD2-046 C O1	7.9885	Adder	9.4
936362	AD2-046 E O1	3.6735	Adder	4.32
936481	AD2-063 C O1	23.8167	50/50	23.8167
936482	AD2-063 E O1	15.7455	50/50	15.7455
937481	AD2-202 C O1	3.1956	50/50	3.1956
937482	AD2-202 E O1	1.7976	50/50	1.7976
938371	AE1-056 C	16.7108	50/50	16.7108
938372	AE1-056 E	9.1306	50/50	9.1306
939181	AE1-148 C O1	8.0059	Adder	9.42
939182	AE1-148 E O1	5.3373	Adder	6.28
940241	AE2-006	0.9343	50/50	0.9343
940661	AE2-053 O1	2.9651	Adder	3.49
941791	AE2-182 C	5.7314	50/50	5.7314
941792	AE2-182 E	2.6681	50/50	2.6681
942451	AE2-258	3.2227	50/50	3.2227
942461	AE2-259 C O1	30.4836	50/50	30.4836
942462	AE2-259 E O1	20.3224	50/50	20.3224
942751	AE2-291 C O1	5.6723	Adder	6.67
942752	AE2-291 E O1	3.7815	Adder	4.45
942761	AE2-292 C O1	7.0625	Adder	8.31
942762	AE2-292 E O1	4.7083	Adder	5.54
943901	AF1-058 C	1.9500	Adder	2.29
943902	AF1-058 E	1.3000	Adder	1.53
943911	AF1-059	13.3237	Adder	15.67
945811	AF1-246 C O1	12.3908	50/50	12.3908
945812	AF1-246 E O1	17.1112	50/50	17.1112
946293	AF1-293 BAT	11.0390	Merchant Transmission	11.0390
946301	AF1-294 C	9.9170	50/50	9.9170
946302	AF1-294 E	6.6113	50/50	6.6113
WEC	WEC	0.2977	Confirmed LTF	0.2977
LGEE	LGEE	0.5625	Confirmed LTF	0.5625
CPL	CPL	3.4947	Confirmed LTF	3.4947
CBM-W2	CBM-W2	12.4079	Confirmed LTF	12.4079

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
NY	NY	0.3882	Confirmed LTF	0.3882
TVA	TVA	2.6446	Confirmed LTF	2.6446
O-066	O-066	5.3962	Confirmed LTF	5.3962
CBM-S2	CBM-S2	23.4495	Confirmed LTF	23.4495
CBM-S1	CBM-S1	14.6629	Confirmed LTF	14.6629
G-007	G-007	0.8414	Confirmed LTF	0.8414
MADISON	MADISON	0.7822	Confirmed LTF	0.7822
MEC	MEC	1.8353	Confirmed LTF	1.8353
AA2-074	AA2-074	2.3814	LTF	2.3814
CBM-W1	CBM-W1	11.0588	Confirmed LTF	11.0588

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
142108023	314692	6FARMVIL	DVP	314677	6BUCKING	DVP	1	DVP_P4-2: 511T556-A	breaker	684.0	106.52	108.92	AC	16.53

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.2285	50/50	0.2285
314429	3JTRSVLE	0.9141	50/50	0.9141
315153	1CLOVER1	23.0754	50/50	23.0754
315154	1CLOVER2	22.8455	50/50	22.8455
924021	AB2-043 C O1	0.7653	50/50	0.7653
924022	AB2-043 E O1	3.8638	50/50	3.8638
924161	AB2-060 C OP	2.2573	50/50	2.2573
924162	AB2-060 E OP	5.9976	50/50	5.9976
924301	AB2-077 C O1 (Suspended)	2.6939	50/50	2.6939
924302	AB2-077 E O1 (Suspended)	1.7959	50/50	1.7959
924311	AB2-078 C O1 (Suspended)	2.6939	50/50	2.6939
924312	AB2-078 E O1 (Suspended)	1.7959	50/50	1.7959
924321	AB2-079 C O1 (Suspended)	2.6939	50/50	2.6939
924322	AB2-079 E O1 (Suspended)	1.7959	50/50	1.7959
925611	AC1-036 C	1.8368	50/50	1.8368
925612	AC1-036 E	2.9969	50/50	2.9969
925781	AC1-054 C O1	5.1752	Adder	6.09
925782	AC1-054 E O1	2.3841	Adder	2.8
925831	AC1-062	0.1357	50/50	0.1357
925991	AC1-075 C	4.2253	Adder	4.97
925992	AC1-075 E	2.3940	Adder	2.82
926021	AC1-080 C	1.4121	Adder	1.66
926022	AC1-080 E	0.7943	Adder	0.93
926271	AC1-105 C O1 (Suspended)	6.7756	50/50	6.7756
926272	AC1-105 E O1 (Suspended)	3.3729	50/50	3.3729
926451	AC1-116 C	-0.3876	Adder	-0.46
927251	AC1-221 C	1.4240	Adder	1.68
927252	AC1-221 E	1.4240	Adder	1.68
927261	AC1-222 C	3.4609	Adder	4.07
927262	AC1-222 E	3.2946	Adder	3.88
932761	AC2-100 C	3.2773	Adder	3.86
932762	AC2-100 E	1.5996	Adder	1.88
934311	AD1-055 C	2.4030	Adder	2.83

Bus #	Bus	Gendoliv MW Impact	Type	Full MW Impact
934312	AD1-055 E	0.6196	Adder	0.73
934341	AD1-058 C	3.5699	Adder	4.2
934342	AD1-058 E	0.9071	Adder	1.07
934611	AD1-087 C O1	12.0620	50/50	12.0620
934612	AD1-087 E O1	5.6689	50/50	5.6689
934621	AD1-088 C	35.2830	50/50	35.2830
934622	AD1-088 E	16.6038	50/50	16.6038
934991	AD1-131 C	1.1704	Adder	1.38
934992	AD1-131 E	0.7803	Adder	0.92
935171	AD1-152 C O1	11.9870	50/50	11.9870
935172	AD1-152 E O1	7.9914	50/50	7.9914
935221	AD1-157 C	3.8768	50/50	3.8768
935222	AD1-157 E	2.5846	50/50	2.5846
936261	AD2-033 C	23.8150	50/50	23.8150
936262	AD2-033 E	15.8766	50/50	15.8766
936361	AD2-046 C O1	7.9917	Adder	9.4
936362	AD2-046 E O1	3.6750	Adder	4.32
936481	AD2-063 C O1	23.8230	50/50	23.8230
936482	AD2-063 E O1	15.7497	50/50	15.7497
937481	AD2-202 C O1	3.1965	50/50	3.1965
937482	AD2-202 E O1	1.7981	50/50	1.7981
938371	AE1-056 C	16.7135	50/50	16.7135
938372	AE1-056 E	9.1321	50/50	9.1321
938561	AE1-075 C	-0.9177	Adder	-1.08
939181	AE1-148 C O1	8.0091	Adder	9.42
939182	AE1-148 E O1	5.3394	Adder	6.28
940241	AE2-006	0.9345	50/50	0.9345
940661	AE2-053 O1	2.9663	Adder	3.49
941791	AE2-182 C	5.7323	50/50	5.7323
941792	AE2-182 E	2.6685	50/50	2.6685
942451	AE2-258	3.2237	50/50	3.2237
942461	AE2-259 C O1	30.4878	50/50	30.4878
942462	AE2-259 E O1	20.3252	50/50	20.3252
942751	AE2-291 C O1	5.6759	Adder	6.68
942752	AE2-291 E O1	3.7839	Adder	4.45
942761	AE2-292 C O1	7.0671	Adder	8.31
942762	AE2-292 E O1	4.7114	Adder	5.54
943901	AF1-058 C	1.9507	Adder	2.29
943902	AF1-058 E	1.3005	Adder	1.53
943911	AF1-059	13.3297	Adder	15.68
945811	AF1-246 C O1	12.3938	50/50	12.3938
945812	AF1-246 E O1	17.1152	50/50	17.1152
946293	AF1-293 BAT	11.0333	Merchant Transmission	11.0333
946301	AF1-294 C	9.9187	50/50	9.9187
946302	AF1-294 E	6.6125	50/50	6.6125
WEC	WEC	0.2999	Confirmed LTF	0.2999
LGEE	LGEE	0.5665	Confirmed LTF	0.5665
CPL	CPL	3.4993	Confirmed LTF	3.4993
CBM-W2	CBM-W2	12.4652	Confirmed LTF	12.4652
NY	NY	0.3843	Confirmed LTF	0.3843
TVA	TVA	2.6544	Confirmed LTF	2.6544
O-066	O-066	5.3491	Confirmed LTF	5.3491

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CBM-S2	CBM-S2	23.4899	Confirmed LTF	23.4899
CBM-S1	CBM-S1	14.7226	Confirmed LTF	14.7226
G-007	G-007	0.8341	Confirmed LTF	0.8341
MADISON	MADISON	0.7822	Confirmed LTF	0.7822
MEC	MEC	1.8464	Confirmed LTF	1.8464
AA2-074	AA2-074	2.3846	LTF	2.3846
CBM-W1	CBM-W1	11.1464	Confirmed LTF	11.1464

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
90243948	313867	6BREMODIST	DVP	313707	6FORK UNION	DVP	1	314333 6POWHATN 230 933500 AC2-165 TAP 230 1	single	661.76	111.53	112.53	AC	6.64

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.1530	80/20	0.1530
314429	3JTRSVLE	0.6120	80/20	0.6120
314572	3EMPORIA	0.0246	80/20	0.0246
314677	6BUCKING	1.2050	80/20	1.2050
314704	3LAWRENC	0.1592	80/20	0.1592
315153	1CLOVER1	8.2207	80/20	8.2207
315154	1CLOVER2	8.1387	80/20	8.1387
315158	1KERR 1	0.2684	80/20	0.2684
315159	1KERR 2	0.7515	80/20	0.7515
315160	1KERR 3	0.7515	80/20	0.7515
315161	1KERR 4	0.7515	80/20	0.7515
315162	1KERR 5	0.7515	80/20	0.7515
315163	1KERR 6	0.7515	80/20	0.7515
315164	1KERR 7	0.7515	80/20	0.7515
315191	1BEARGRDN G1	17.0060	80/20	17.0060
315192	1BEARGRDN G2	17.0060	80/20	17.0060
315193	1BEARGRDN S1	34.9994	80/20	34.9994
315266	1PLYWOOD A	0.7542	80/20	0.7542
920291	AA2-127	3.5329	80/20	3.5329
923572	AB1-173 C OP	0.7321	80/20	0.7321
923582	AB1-173AC OP	0.7321	80/20	0.7321
923911	AB2-031 C O1	0.7267	80/20	0.7267
923991	AB2-040 C O1 (Suspended)	2.3861	80/20	2.3861
924021	AB2-043 C O1	0.4798	80/20	0.4798
924161	AB2-060 C OP	1.4204	80/20	1.4204
924301	AB2-077 C O1 (Suspended)	1.6786	80/20	1.6786
924311	AB2-078 C O1 (Suspended)	1.6786	80/20	1.6786
924321	AB2-079 C O1 (Suspended)	1.6786	80/20	1.6786
925171	AB2-174 C O1	0.4072	80/20	0.4072
925611	AC1-036 C	1.2041	80/20	1.2041
925781	AC1-054 C O1	3.9467	80/20	3.9467
925831	AC1-062	0.0908	80/20	0.0908
925991	AC1-075 C	2.3861	80/20	2.3861
926021	AC1-080 C	0.7974	80/20	0.7974
926271	AC1-105 C O1 (Suspended)	3.3665	80/20	3.3665

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
926451	AC1-116 C	-0.7534	Adder	-0.89
927251	AC1-221 C	0.7900	80/20	0.7900
927261	AC1-222 C	1.8732	80/20	1.8732
932511	AC2-071 C	6.7029	80/20	6.7029
932761	AC2-100 C	1.8181	80/20	1.8181
933501	AC2-165 C	35.3092	80/20	35.3092
934311	AD1-055 C	1.3006	80/20	1.3006
934341	AD1-058 C	1.9804	80/20	1.9804
934611	AD1-087 C O1	4.5412	80/20	4.5412
934621	AD1-088 C	21.9119	80/20	21.9119
934991	AD1-131 C	0.6493	80/20	0.6493
935171	AD1-152 C O1	4.5130	80/20	4.5130
935221	AD1-157 C	2.6094	80/20	2.6094
936261	AD2-033 C	15.5181	80/20	15.5181
936361	AD2-046 C O1	5.8992	80/20	5.8992
936481	AD2-063 C O1	15.2550	80/20	15.2550
937481	AD2-202 C O1	1.2035	80/20	1.2035
938371	AE1-056 C	11.2493	80/20	11.2493
938821	AE1-108 C	11.5943	80/20	11.5943
939181	AE1-148 C O1	5.9114	80/20	5.9114
940241	AE2-006	0.6126	80/20	0.6126
940661	AE2-053 O1	2.1894	80/20	2.1894
941791	AE2-182 C	3.5599	80/20	3.5599
942451	AE2-258	2.0211	80/20	2.0211
942461	AE2-259 C O1	20.7630	80/20	20.7630
942751	AE2-291 C O1	3.2987	80/20	3.2987
942761	AE2-292 C O1	4.1072	80/20	4.1072
943901	AF1-058 C	1.0472	80/20	1.0472
943911	AF1-059	10.4290	80/20	10.4290
945811	AF1-246 C O1	3.5692	80/20	3.5692
946281	AF1-292 C	0.4991	80/20	0.4991
946291	AF1-293 C O1	6.7642	80/20	6.7642
946301	AF1-294 C	6.6408	80/20	6.6408
946591	AF1-323 C	3.7300	80/20	3.7300
WEC	WEC	0.0608	Confirmed LTF	0.0608
LGEE	LGEE	0.1227	Confirmed LTF	0.1227
CPL	CPL	2.2273	Confirmed LTF	2.2273
CBM-W2	CBM-W2	4.9877	Confirmed LTF	4.9877
NY	NY	0.4048	Confirmed LTF	0.4048
TVA	TVA	1.2278	Confirmed LTF	1.2278
CBM-S2	CBM-S2	14.2361	Confirmed LTF	14.2361
CBM-S1	CBM-S1	6.4241	Confirmed LTF	6.4241
MADISON	MADISON	0.5806	Confirmed LTF	0.5806
MEC	MEC	0.5498	Confirmed LTF	0.5498
AA2-074	AA2-074	1.5152	LTF	1.5152
CBM-W1	CBM-W1	1.9265	Confirmed LTF	1.9265

11.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42959331	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P7-1: LN 22-90-A	tower	199.0	123.95	126.02	AC	4.89

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
315158	1KERR 1	0.9286	50/50	0.9286
315159	1KERR 2	2.6001	50/50	2.6001
315160	1KERR 3	2.6001	50/50	2.6001
315161	1KERR 4	2.6001	50/50	2.6001
315162	1KERR 5	2.6001	50/50	2.6001
315163	1KERR 6	2.6001	50/50	2.6001
315164	1KERR 7	2.6001	50/50	2.6001
924021	AB2-043 C O1	0.6513	50/50	0.6513
924022	AB2-043 E O1	3.2884	50/50	3.2884
924161	AB2-060 C OP	1.8513	50/50	1.8513
924162	AB2-060 E OP	4.9188	50/50	4.9188
924301	AB2-077 C O1 (Suspended)	2.3768	50/50	2.3768
924302	AB2-077 E O1 (Suspended)	1.5846	50/50	1.5846
924311	AB2-078 C O1 (Suspended)	2.3768	50/50	2.3768
924312	AB2-078 E O1 (Suspended)	1.5846	50/50	1.5846
924321	AB2-079 C O1 (Suspended)	2.3768	50/50	2.3768
924322	AB2-079 E O1 (Suspended)	1.5846	50/50	1.5846
925611	AC1-036 C	0.8792	50/50	0.8792
925612	AC1-036 E	1.4345	50/50	1.4345
925781	AC1-054 C O1	15.5545	50/50	15.5545
925782	AC1-054 E O1	7.1656	50/50	7.1656
926271	AC1-105 C O1 (Suspended)	2.4425	Adder	2.87
926272	AC1-105 E O1 (Suspended)	1.2158	Adder	1.43
935221	AD1-157 C	0.8225	Adder	0.97
935222	AD1-157 E	0.5483	Adder	0.65
936261	AD2-033 C	12.6001	50/50	12.6001
936262	AD2-033 E	8.4001	50/50	8.4001
936361	AD2-046 C O1	17.8264	50/50	17.8264
936362	AD2-046 E O1	8.1976	50/50	8.1976
936481	AD2-063 C O1	16.1136	50/50	16.1136
936482	AD2-063 E O1	10.6529	50/50	10.6529
938371	AE1-056 C	3.5457	Adder	4.17
938372	AE1-056 E	1.9373	Adder	2.28
939181	AE1-148 C O1	17.2071	50/50	17.2071

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939182	AE1-148 E O1	11.4714	50/50	11.4714
940241	AE2-006	0.4473	50/50	0.4473
940661	AE2-053 O1	6.3730	50/50	6.3730
942451	AE2-258	2.7436	50/50	2.7436
942461	AE2-259 C O1	3.7837	Adder	4.45
942462	AE2-259 E O1	2.5225	Adder	2.97
943901	AF1-058 C	0.5678	Adder	0.67
943902	AF1-058 E	0.3786	Adder	0.45
943911	AF1-059	12.3240	50/50	12.3240
946301	AF1-294 C	2.4962	Adder	2.94
946302	AF1-294 E	1.6642	Adder	1.96
NEWTON	NEWTON	0.4491	Confirmed LTF	0.4491
FARMERCITY	FARMERCITY	0.0260	Confirmed LTF	0.0260
G-007A	G-007A	0.1630	Confirmed LTF	0.1630
VFT	VFT	0.4321	Confirmed LTF	0.4321
PRAIRIE	PRAIRIE	1.2269	Confirmed LTF	1.2269
COFFEEN	COFFEEN	0.0840	Confirmed LTF	0.0840
CHEOAH	CHEOAH	0.3819	Confirmed LTF	0.3819
EDWARDS	EDWARDS	0.1351	Confirmed LTF	0.1351
TILTON	TILTON	0.2381	Confirmed LTF	0.2381
GIBSON	GIBSON	0.2157	Confirmed LTF	0.2157
CALDERWOOD	CALDERWOOD	0.3718	Confirmed LTF	0.3718
BLUEG	BLUEG	0.6614	Confirmed LTF	0.6614
TRIMBLE	TRIMBLE	0.2098	Confirmed LTF	0.2098
CATAWBA	CATAWBA	0.4196	Confirmed LTF	0.4196

11.7.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
90243928	314747	6BREMO	DVP	313867	6BREMODIST	DVP	1	314333 6POWHATN 230 933500 AC2-165 TAP 230 1	single	661.76	113.82	114.82	AC	6.64

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.1530	80/20	0.1530
314429	3JTRSVLE	0.6120	80/20	0.6120
314572	3EMPORIA	0.0246	80/20	0.0246
314677	6BUCKING	1.2050	80/20	1.2050
314704	3LAWRENC	0.1592	80/20	0.1592
315153	1CLOVER1	8.2207	80/20	8.2207
315154	1CLOVER2	8.1387	80/20	8.1387
315158	1KERR 1	0.2684	80/20	0.2684
315159	1KERR 2	0.7515	80/20	0.7515
315160	1KERR 3	0.7515	80/20	0.7515
315161	1KERR 4	0.7515	80/20	0.7515
315162	1KERR 5	0.7515	80/20	0.7515
315163	1KERR 6	0.7515	80/20	0.7515
315164	1KERR 7	0.7515	80/20	0.7515
315191	1BEARGRDN G1	17.0060	80/20	17.0060
315192	1BEARGRDN G2	17.0060	80/20	17.0060
315193	1BEARGRDN S1	34.9994	80/20	34.9994
315266	1PLYWOOD A	0.7542	80/20	0.7542
920291	AA2-127	3.5329	80/20	3.5329
923572	AB1-173 C OP	0.7321	80/20	0.7321
923582	AB1-173AC OP	0.7321	80/20	0.7321
923911	AB2-031 C O1	0.7267	80/20	0.7267
923991	AB2-040 C O1 (Suspended)	2.3861	80/20	2.3861
924021	AB2-043 C O1	0.4798	80/20	0.4798
924161	AB2-060 C OP	1.4204	80/20	1.4204
924301	AB2-077 C O1 (Suspended)	1.6786	80/20	1.6786
924311	AB2-078 C O1 (Suspended)	1.6786	80/20	1.6786
924321	AB2-079 C O1 (Suspended)	1.6786	80/20	1.6786
925171	AB2-174 C O1	0.4072	80/20	0.4072
925611	AC1-036 C	1.2041	80/20	1.2041
925781	AC1-054 C O1	3.9467	80/20	3.9467
925831	AC1-062	0.0908	80/20	0.0908
925991	AC1-075 C	2.3861	80/20	2.3861
926021	AC1-080 C	0.7974	80/20	0.7974
926271	AC1-105 C O1 (Suspended)	3.3665	80/20	3.3665

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
926451	AC1-116 C	-0.7534	Adder	-0.89
927251	AC1-221 C	0.7900	80/20	0.7900
927261	AC1-222 C	1.8732	80/20	1.8732
932511	AC2-071 C	6.7029	80/20	6.7029
932761	AC2-100 C	1.8181	80/20	1.8181
933501	AC2-165 C	35.3092	80/20	35.3092
934311	AD1-055 C	1.3006	80/20	1.3006
934341	AD1-058 C	1.9804	80/20	1.9804
934611	AD1-087 C O1	4.5412	80/20	4.5412
934621	AD1-088 C	21.9119	80/20	21.9119
934991	AD1-131 C	0.6493	80/20	0.6493
935171	AD1-152 C O1	4.5130	80/20	4.5130
935221	AD1-157 C	2.6094	80/20	2.6094
936261	AD2-033 C	15.5181	80/20	15.5181
936361	AD2-046 C O1	5.8992	80/20	5.8992
936481	AD2-063 C O1	15.2550	80/20	15.2550
937481	AD2-202 C O1	1.2035	80/20	1.2035
938371	AE1-056 C	11.2493	80/20	11.2493
938821	AE1-108 C	11.5943	80/20	11.5943
939181	AE1-148 C O1	5.9114	80/20	5.9114
940241	AE2-006	0.6126	80/20	0.6126
940661	AE2-053 O1	2.1894	80/20	2.1894
941791	AE2-182 C	3.5599	80/20	3.5599
942451	AE2-258	2.0211	80/20	2.0211
942461	AE2-259 C O1	20.7630	80/20	20.7630
942751	AE2-291 C O1	3.2987	80/20	3.2987
942761	AE2-292 C O1	4.1072	80/20	4.1072
943901	AF1-058 C	1.0472	80/20	1.0472
943911	AF1-059	10.4290	80/20	10.4290
945811	AF1-246 C O1	3.5692	80/20	3.5692
946281	AF1-292 C	0.4991	80/20	0.4991
946291	AF1-293 C O1	6.7642	80/20	6.7642
946301	AF1-294 C	6.6408	80/20	6.6408
946591	AF1-323 C	3.7300	80/20	3.7300
WEC	WEC	0.0608	Confirmed LTF	0.0608
LGEE	LGEE	0.1227	Confirmed LTF	0.1227
CPL	CPL	2.2273	Confirmed LTF	2.2273
CBM-W2	CBM-W2	4.9877	Confirmed LTF	4.9877
NY	NY	0.4048	Confirmed LTF	0.4048
TVA	TVA	1.2278	Confirmed LTF	1.2278
CBM-S2	CBM-S2	14.2361	Confirmed LTF	14.2361
CBM-S1	CBM-S1	6.4241	Confirmed LTF	6.4241
MADISON	MADISON	0.5806	Confirmed LTF	0.5806
MEC	MEC	0.5498	Confirmed LTF	0.5498
AA2-074	AA2-074	1.5152	LTF	1.5152
CBM-W1	CBM-W1	1.9265	Confirmed LTF	1.9265

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA2-074	CPLP-PJM	Confirmed
AA2-127	Bear Garden 230kV	In Service
AB1-173	Brink-Trego 115kV	Engineering and Procurement
AB1-173A	Brink-Trego 115kV	Engineering and Procurement
AB2-031	Brink-Trego 115kV	Engineering and Procurement
AB2-040	Brink 115kV	Suspended
AB2-043	Chase City 115kV	Under Construction
AB2-045	Buckingham 34.5kV	In Service
AB2-060	Chase City-Lunenburg 115kV	Under Construction
AB2-077	Buggs Island-Chase City 115kV	Suspended
AB2-078	Buggs Island-Chase City 115kV	Suspended
AB2-079	Buggs Island-Chase City 115kV	Suspended
AB2-174	Emporia-Trego 115kV	Under Construction
AC1-036	Twittys Creek 34.5kV	Engineering and Procurement
AC1-054	Kerr Dam-Eatons Ferry 115 kV	Engineering and Procurement
AC1-062	Jetersville 34.5kV	In Service
AC1-075	Perth-Hickory Grove 115kV	Engineering and Procurement
AC1-080	Perth-Hickory Grove 115kV	Engineering and Procurement
AC1-105	Halifax-Mt. Laurel 115kV	Suspended
AC1-116	Mount Eagle 34.5kV	Engineering and Procurement
AC1-221	Halifax-Person 230kV	Engineering and Procurement
AC1-222	Crystal Hill-Halifax 115kV	Engineering and Procurement
AC2-071	Buckingham 35kV	Engineering and Procurement
AC2-100	Halifax-Person 230kV	Active
AC2-165	Bremo-Powhatan 230kV	Active
AD1-055	Crystal Hill-Halifax 115 kV	Engineering and Procurement
AD1-058	Halifax-Person 230 kV	Engineering and Procurement
AD1-087	Clover-Sedge Hill 230 kV	Active
AD1-088	Briery-Clover 230 kV	Active
AD1-131	Sedge Hill-Person 230 kV	Active
AD1-152	Clover-Sedge Hill 230 kV	Active
AD1-157	South Creek 34.5 kV	Engineering and Procurement
AD2-033	Chase City-Lunenburg 115 kV	Active
AD2-046	Boydton DP-Kerr Dam 115 kV	Active
AD2-063	Central-Chase City 115kV	Active
AD2-202	Clover-Sedge Hill 230kV	Active
AE1-056	Red House-South Creek 115 kV	Active
AE1-075	Powhatan 34.5 kV	Engineering and Procurement
AE1-108	Bremo-Scottsville 138 kV	Active

Queue Number	Project Name	Status
AE1-148	Kerr Dam-Ridge Rd 115 kV	Active
AE2-006	Twittys Creek 34.5 kV	Engineering and Procurement
AE2-053	Kerr Dam-Ridge Road 115 kV	Active
AE2-182	Briery-Clover 230 kV	Active
AE2-258	Chase City 115 kV	Active
AE2-259	Curdsville-Willis Mtn 115 kV	Active
AE2-291	Grit DP-Perth 115 kV	Active
AE2-292	Grit DP-Perth 115 kV	Active
AF1-058	Welco 34.5 kV	Engineering and Procurement
AF1-059	Brodnax-South Hill 115 kV	Active
AF1-246	Clover-Rawlings 500 kV	Active
AF1-292	Fields Crossroads 34.5 kV	Active
AF1-293	Kidds Store-Fort Union 115 kV	Active
AF1-294	Jetersville-Ponton 115 kV	Active
AF1-323	Scottsville-Colleen 138 kV	Active

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
DVP_P4-2: 102802	CONTINGENCY 'DVP_P4-2: 102802' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P4-6: CAROLIN T122	CONTINGENCY 'DVP_P4-6: CAROLIN T122' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 314559 TO BUS 315126 CKT 1 /* 3CAROLNA 115.00 - 1ROARAP2 14.400 OPEN BRANCH FROM BUS 314559 TO BUS 315128 CKT 1 /* 3CAROLNA 115.00 - 1ROARAP4 14.400 OPEN BUS 315126 /* ISLAND: 1ROARAP2 14.400 OPEN BUS 315128 /* ISLAND: 1ROARAP4 14.400 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314578 CKT 1 /* 3CAROLNA 115.00 - 3HORNRTN 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 OPEN BUS 314559 /* 3CAROLNA 115.00 KV END
DVP_P1-2: LN 556-A	CONTINGENCY 'DVP_P1-2: LN 556-A' OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 1 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 2 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 3 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314906 TO BUS 945810 CKT 1 /* 8CLOVER 500.00 - AF1-246 TAP 500.00 OPEN BUS 314906 /* ISLAND: 8CLOVER 500.00 OPEN BUS 314915 /* ISLAND: 8CLOVER_STC 500.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 556-B	CONTINGENCY 'DVP_P1-2: LN 556-B' OPEN BRANCH FROM BUS 945810 TO BUS 314936 CKT 1 /* AF1-246 TAP 500.00 - 8RAWLINGS 500.00 END
DVP_P7-1: LN 22-90-A	CONTINGENCY 'DVP_P7-1: LN 22-90-A' OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1 /* 3EATON F 115.00 - AC1- 054 TAP 115.00 OPEN BUS 314571 /* ISLAND: 3EATON F 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1 /* 3FIVEFORKSDP115.00 - 3LITTLTN 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1 /* 3FIVEFORKSDP115.00 - 3PALMERSPRNG115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1 /* 3LITTLTN 115.00 - 3L GASTN 115.00 OPEN BUS 314265 /* ISLAND: 3FIVEFORKSDP115.00 OPEN BUS 314584 /* ISLAND: 3LITTLTN 115.00 OPEN BUS 314585 /* ISLAND: 3L GASTN 115.00 END
DVP_P4-2: 556T591	CONTINGENCY 'DVP_P4-2: 556T591' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 945810 TO BUS 314936 CKT 1 /* AF1-246 TAP 500.00 - 8RAWLINGS 500.00 OPEN BRANCH FROM BUS 314935 TO BUS 314936 CKT 1 /* 8HERITAGE 500.00 - 8RAWLINGS 500.00 END
DVP_P1-2: LN 98-C	CONTINGENCY 'DVP_P1-2: LN 98-C' OPEN BRANCH FROM BUS 924160 TO BUS 314267 CKT 1 /* AB2-060 TAP 115.00 - 3CHASCTY2 115.00 END
DVP_P7-1: LN 22-90-B	CONTINGENCY 'DVP_P7-1: LN 22-90-B' OPEN BRANCH FROM BUS 925780 TO BUS 314702 CKT 1 /* AC1-054 TAP 115.00 - 3KERR 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1 /* 3FIVEFORKSDP115.00 - 3LITTLTN 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1 /* 3FIVEFORKSDP115.00 - 3PALMERSPRNG115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1 /* 3LITTLTN 115.00 - 3L GASTN 115.00 OPEN BUS 314265 /* ISLAND: 3FIVEFORKSDP115.00 OPEN BUS 314584 /* ISLAND: 3LITTLTN 115.00 OPEN BUS 314585 /* ISLAND: 3L GASTN 115.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 2027-B	CONTINGENCY 'DVP_P1-2: LN 2027-B' OPEN BRANCH FROM BUS 933500 TO BUS 314333 CKT 1 /* AC2-165 TAP 230.00 - 6POWHATN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314322 CKT 1 /* 6JUDES F 230.00 - 6MDLTHAN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314333 CKT 1 /* 6JUDES F 230.00 - 6POWHATN 230.00 OPEN BUS 314310 /* ISLAND: 6JUDES F 230.00 OPEN BUS 314333 /* ISLAND: 6POWHATN 230.00 OPEN BUS 923862 /* ISLAND: AB2-026 E 230.00 END
DVP_P4-2: 101002	CONTINGENCY 'DVP_P4-2: 101002' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 313723 TO BUS 314259 CKT 1 /* 3PECAN 115.00 - 3CAROL56_1 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314835 CKT 1 /* 3CAROL56_1 115.00 - 3CAROL_1 115.00 OPEN BUS 314259 /* ISLAND: 3CAROL56_1 115.00 OPEN BUS 314835 /* ISLAND: 3CAROL_1 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P4-2: 2202	CONTINGENCY 'DVP_P4-2: 2202' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1 /* 3EATON F 115.00 - AC1-054 TAP 115.00 OPEN BUS 314571 /* ISLAND: 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P1-2: LN 296-A	CONTINGENCY 'DVP_P1-2: LN 296-A' OPEN BRANCH FROM BUS 304070 TO BUS 927250 CKT 1 /* 6PERSON230 T230.00 - AC1-221 TAP 230.00 END

Contingency Name	Contingency Definition
DVP_P4-2: 13002	CONTINGENCY 'DVP_P4-2: 13002' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314595 TO BUS 314600 CKT 1 /* 3PL HILL 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314600 TO BUS 314615 CKT 1 /* 3PLHITP 115.00 - 3SKIPPERS 115.00 OPEN BUS 314595 /* ISLAND: 3PL HILL 115.00 OPEN BUS 314600 /* ISLAND: 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P1-2: LN 296-B	CONTINGENCY 'DVP_P1-2: LN 296-B' OPEN BRANCH FROM BUS 927250 TO BUS 314697 CKT 1 /* AC1-221 TAP T230.00 - 6SEdge HILL 230.00 END
314333 6POWHATN 230 933500 AC2-165 TAP 230 1	CONTINGENCY '314333 6POWHATN 230 933500 AC2-165 TAP 230 1' OPEN BRANCH FROM BUS 314333 TO BUS 933500 CKT 1 END
DVP_P4-2: 511T556-A	CONTINGENCY 'DVP_P4-2: 511T556-A' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 940470 TO BUS 314936 CKT 1 /* AE2-031 TAP 500.00 - 8RAWLINGS 500.00 OPEN BRANCH FROM BUS 945810 TO BUS 314936 CKT 1 /* AF1-246 TAP 500.00 - 8RAWLINGS 500.00 END
DVP_P4-2: 511T591	CONTINGENCY 'DVP_P4-2: 511T591' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 940470 TO BUS 314936 CKT 1 /* AE2-031 TAP 500.00 - 8RAWLINGS 500.00 OPEN BRANCH FROM BUS 314935 TO BUS 314936 CKT 1 /* 8HERITAGE 500.00 - 8RAWLINGS 500.00 END

12 Light Load Analysis

Not required for this project.

13 Short Circuit Analysis

No circuit breakers were identified as overdutied as part of this analysis.

14 Stability and Reactive Power

To be determined in the Facilities Study Phase.

15 Affected Systems

15.1 Duke Energy-Progress

Potential constraints were identified by PJM on the following Dominion – Duke Energy/Progress (DEP) tie lines. Enter into an Affected System Facilities Study agreement with Duke / Progress Energy (DEP) to determine what, if any, reinforcements are required on their system. The following facilities were identified in this report:

- Kerr Dam – GW King Tap 115 kV line

Attachment 1: One Line Diagram

